A First-Principles Primer: Wavelet Basis Sets and the Road from a Molecule to a Qubit Hamiltonian

Basis sets, Daubechies wavelets, electron integrals, and fermion-to-qubit mappings

Companion notes for the NTU wavelet-QC deck and demo
(a refresher, written to fill the gaps the slides assume)

18 July 2026

Abstract

These notes explain, from first principles, the machinery the NTU wavelet-QC deck uses but does not derive: what a *basis set* is and why its size is a quantum computer's qubit budget; what a *Daubechies wavelet* is and why a real-space wavelet basis is systematically improvable and linear-scaling; how the one- and two-electron integrals h_{pq} and h_{pqrs} are actually computed in BIGDFT (including the Poisson-solver trick for the two-electron term); and how a second-quantized molecular Hamiltonian is mapped onto qubits by the Jordan–Wigner, parity, and Bravyi–Kitaev transforms. Each part is anchored by a figure from the deck and points at the module in the companion `wavelet-qc-demo` codebase that implements it. The running example is the same one the demo validates: the nitrogen molecule in a Daubechies-wavelet basis. This is a teaching document, not a paper.

Contents

1 Roadmap: one molecule, four questions **3**

2 Part A — What is a basis set? **4**
2.1 The problem: an equation in $3N$ variables . . . 4
2.2 One-particle orbitals and the LCAO expansion . . . 4
2.3 The integrals: what a Hamiltonian is made of . . . 4
2.4 Gaussian basis sets and the accuracy hierarchy . . . 4

3 Part B — Wavelets from first principles **5**
3.1 Why not just refine a Gaussian basis? . . . 5
3.2 Multiresolution analysis . . . 5
3.3 The Daubechies family . . . 6
3.4 Recovering φ and ψ: the cascade algorithm . . . 6

4 Part C — Wavelets in BigDFT and computing the integrals **7**
4.1 The adaptive grid . . . 7
4.2 From grid to orbitals: Kohn–Sham DFT with pseudopotentials . . . 7
4.3 The two-electron integral: the Poisson-solver trick . . . 8
4.4 Symmetries and locality . . . 8

5 Part D — From a Hamiltonian to qubits **10**
5.1 Second quantization . . . 10
5.2 The mapping problem . . . 10
5.3 Jordan–Wigner . . . 10
5.4 Parity, tapering, and Bravyi–Kitaev . . . 10
5.5 Qubit-efficient encodings . . . 11

6 Part E — Solving on a quantum computer **12**
6.1 Active-space selection . . . 12
6.2 The variational quantum eigensolver . . . 12
6.3 QSCI / SQD . . . 12

7 Making the wavelet method comparable **13**
7.1 What is comparable . . . 14
7.2 The matched-pseudopotential reference . . . 14
7.3 The result . . . 14

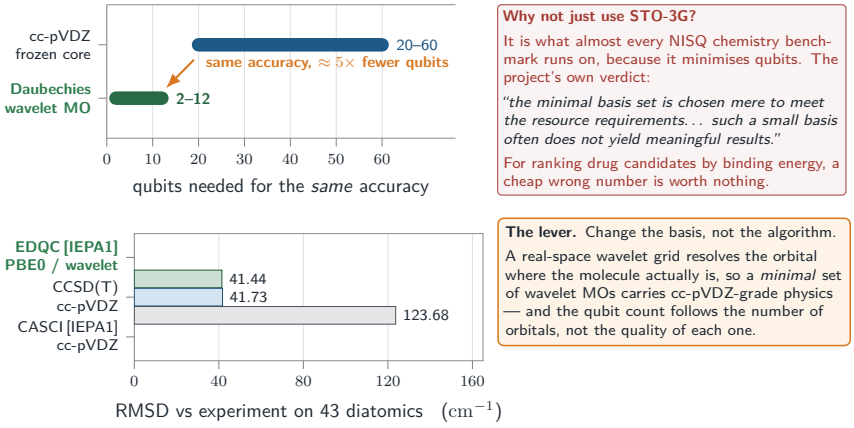
8 How this maps to the demo **15**

9 Curated reading path **16**

10 Glossary **16**

1 Roadmap: one molecule, four questions

The deck makes a single claim: *the basis set is the qubit budget.* A minimal Gaussian basis (STO-3G) is cheap in qubits but too inaccurate to rank drug candidates; a good Gaussian basis (cc-pVDZ) is accurate but needs 20–60 qubits for a small molecule; Daubechies-wavelet molecular orbitals from BIGDFT deliver cc-pVDZ-grade accuracy at 2–12 qubits [1, 2].



Published results of the group this work builds on: Hong *et al.*, PRX Quantum **3**, 020360 (2022); Chou *et al.*, Phys. Rev. Research **5**, 043216 (2023).

Figure 1: The thesis of the whole project, and of these notes: at equal accuracy, a minimal wavelet basis needs a fraction of the qubits a Gaussian basis does. Numbers are the group's published benchmarks [1, 2]. (Reproduced from the deck.)

To understand that claim you need four things the deck takes for granted. This primer is one part per question:

Part A — What is a basis set? How a molecule's wavefunction becomes a finite list of numbers, and why that list's length is the qubit count.

Part B — What is a wavelet? Why a real-space, multiresolution basis is systematically improvable where a fixed small Gaussian basis is not.

Part C — Where do h_{pq} and h_{pqrs} come from? How BIGDFT turns wavelet orbitals into the integrals a Hamiltonian is built from.

Part D — How does a Hamiltonian become qubits? Second quantization and the Jordan–Wigner, parity, and Bravyi–Kitaev mappings.

Part E then closes the loop: how VQE and SQD extract an energy from the qubit Hamiltonian. A final section maps every concept onto a file in the runnable `wavelet-qc-demo`, which reproduces the numbers below from the existing integral files.

Key idea

The through-line of these notes is a pair of integrals. Everything in Parts A–C exists to compute the one-electron integral h_{pq} and the two-electron integral h_{pqrs}; everything in Parts D–E exists to turn those numbers into an energy on a quantum computer. Keep your eye on h_{pq} and h_{pqrs} and the whole pipeline stays legible.

2 Part A — What is a basis set?

2.1 The problem: an equation in $3N$ variables

After the Born–Oppenheimer approximation fixes the nuclei, the electronic Hamiltonian for N electrons is (atomic units)

$$\hat{H} = -\sum_i \tfrac{1}{2}\nabla_i^2 - \sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}. \tag{1}$$

The ground-state wavefunction $\Psi(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ lives in a $3N$-dimensional space (plus spin). We cannot store a function of $3N$ continuous variables, so we expand it in a finite set of building blocks. That set is the **basis set**, and the whole cost of the calculation flows from its size.

2.2 One-particle orbitals and the LCAO expansion

The first reduction is to build the many-electron Ψ out of one-electron functions — *molecular orbitals* (MOs) $\psi_p(\mathbf{r})$ — and to expand each MO in a fixed set of K *atomic-orbital* (AO) basis functions $\{\chi_\mu\}$:

$$\psi_p(\mathbf{r}) = \sum_{\mu=1}^{K} C_{\mu p}\, \chi_\mu(\mathbf{r}). \tag{2}$$

This is the "linear combination of atomic orbitals" (LCAO) idea. A Hartree–Fock (HF) or Kohn–Sham (KS) calculation finds the coefficients $C_{\mu p}$; the number of MOs equals the number of AO functions K. K **is the number that becomes the qubit count** — each spatial MO carries two spin-orbitals, and (before any reduction) each spin-orbital is one qubit.

2.3 The integrals: what a Hamiltonian is made of

In the MO basis, the electronic Hamiltonian (1) is completely specified by two tables of numbers — the one- and two-electron integrals:

$$h_{pq} = \int d\mathbf{x}\; \psi_p^*(\mathbf{x}) \Big(-\tfrac{1}{2}\nabla^2 - \sum_A \tfrac{Z_A}{|\mathbf{r}-\mathbf{R}_A|} \Big) \psi_q(\mathbf{x}), \tag{3}$$

$$h_{pqrs} = \int d\mathbf{x}_1\, d\mathbf{x}_2\; \frac{\psi_p^*(\mathbf{x}_1)\psi_q(\mathbf{x}_1)\, \psi_r^*(\mathbf{x}_2)\psi_s(\mathbf{x}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|} \;\equiv\; [pq|rs]. \tag{4}$$

h_{pq} collects the kinetic energy and the electron–nucleus attraction; h_{pqrs} is the electron–electron repulsion, written here in *chemist* notation $[pq|rs]$. These are exactly the quantities Part C computes and Part D turns into qubits. Note the scaling: there are $\mathcal{O}(K^2)$ one-electron integrals and $\mathcal{O}(K^4)$ two-electron integrals, so the basis size drives cost twice over — once in the qubit count, once in the integral count.

2.4 Gaussian basis sets and the accuracy hierarchy

Almost all quantum chemistry uses *Gaussian-type orbitals* (GTOs), $\chi(\mathbf{r}) \sim x^a y^b z^c e^{-\alpha r^2}$, because the two-electron integrals (4) then have closed forms. A real atomic orbital is better modelled by a Slater function $e^{-\zeta r}$, so several Gaussians are *contracted* to imitate one Slater orbital. The families form an accuracy ladder:

- **Minimal (STO-3G)** [3]: one contracted function per occupied atomic orbital, each a fixed sum of three Gaussians. Smallest possible K — and the reason it is the default in near-term quantum-computing benchmarks — but too crude for quantitative energetics.
- **Split-valence (6-31G, 6-31G*)** [4, 5]: valence orbitals split into tight+loose parts, optionally with polarization (d) functions. A workhorse for organic chemistry.
- **Correlation-consistent (cc-pVXZ, X = D, T, Q, . . .)** [6]: designed so that adding the next "zeta" shell recovers a predictable increment of correlation energy, giving a systematic route to the complete-basis-set (CBS) limit. cc-pVDZ is the smallest member; it is the accuracy target the wavelet work matches.

Basis	Size for N_2 (spatial)	Qubits (JW)	Character
STO-3G	10	20	minimal; cheap, inaccurate
6-31G	18	36	split valence
cc-pVDZ	28	56	first correlation-consistent rung
Wavelet MO (this work)	5–15	10–30→**2–12**	real-space, then MP2-reduced

Common confusion

"Minimal basis" does *not* mean "good enough small model". STO-3G is chosen in near-term quantum-computing papers because it minimises qubits, not because it is accurate; the project's own verdict is that "such a small basis often does not yield meaningful results." A concrete cautionary case is the STO-3G quantum-computing study of protein–ligand binding by Kirsopp et al. [7], whose ligand ranking suffered from exactly this low-quality Hamiltonian. The wavelet route keeps the qubit count minimal while replacing the low-quality Hamiltonian — accuracy comes from a *better* minimal basis, not a bigger one.

3 Part B — Wavelets from first principles

3.1 Why not just refine a Gaussian basis?

A fixed small Gaussian basis is not *systematically* improvable: to get closer to the truth you must swap the whole basis for a larger, differently-optimised one, and the qubit count balloons with it. A real-space basis — represent the orbital by its values on a grid — is systematically improvable (just refine the grid) but a uniform fine grid is astronomically expensive in vacuum. Wavelets resolve this tension: they are a real-space basis that refines *locally*, only where the wavefunction has structure.

3.2 Multiresolution analysis

A *multiresolution analysis* is a ladder of nested approximation spaces

$$\cdots \subset V_{-1} \subset V_0 \subset V_1 \subset \cdots, \qquad \overline{\bigcup_j V_j} = L^2(\mathbb{R}), \tag{5}$$

where V_j is spanned by integer translates of a single *scaling function* φ dilated to resolution 2^j: $\varphi_{j,k}(x) = 2^{j/2}\varphi(2^j x - k)$ [8]. Because $V_0 \subset V_1$, the scaling function at coarse resolution must be expressible in the finer one — the **refinement (dilation) equation**:

$$\varphi(x) = \sqrt{2}\sum_k h_k\, \varphi(2x - k). \tag{6}$$

The *detail* lost between V_j and V_{j+1} lives in a complementary space W_j spanned by translates of a *wavelet* ψ, built from the same coefficients with alternating signs:

$$\psi(x) = \sqrt{2}\sum_k g_k\,\varphi(2x-k), \qquad g_k = (-1)^k h_{1-k}. \tag{7}$$

So an orbital is stored as coarse *scaling-function* coefficients everywhere plus *wavelet* coefficients wherever extra resolution is needed — refinement is local and adaptive.

3.3 The Daubechies family

Daubechies constructed the filters $\{h_k\}$ that make φ, ψ simultaneously [9]:

- **orthonormal** — $\int \varphi(x)\varphi(x-l)\,dx = \delta_{l0}$, which requires $\sum_k h_k h_{k+2l} = \delta_{l0}$;
- **compactly supported** — only $2m$ coefficients are nonzero, so φ vanishes outside an interval of width $2m-1$ (the family is labelled by its order $2m$; BIGDFT uses $2m = 16$);
- **smooth, with m vanishing moments** — $\int x^n \psi(x)\,dx = 0$ for $n < m$, so the basis represents polynomials up to degree $m-1$ exactly (order 16 $\Rightarrow$ exact through 7th order).

Compact support is the property that matters most downstream: because each basis function is nonzero only over a few grid spacings, the overlap of two well-separated orbitals is *exactly* zero, which makes operators sparse and integrals local — the seed of linear scaling (Part C).

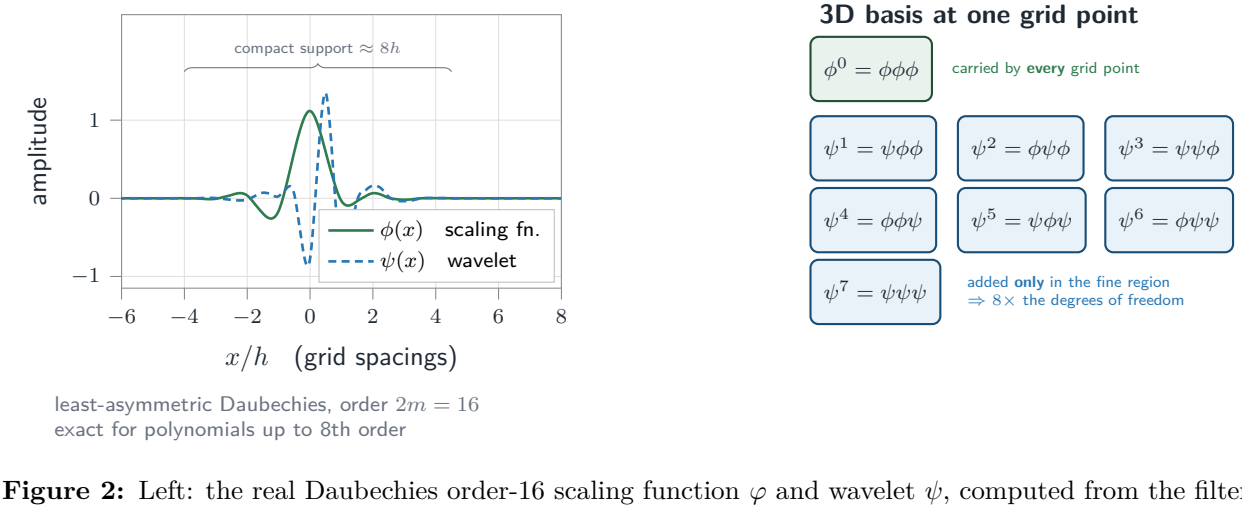


Figure 2: Left: the real Daubechies order-16 scaling function φ and wavelet ψ, computed from the filter $\{h_k\}$ by the cascade algorithm (there is no closed form). Right: in 3D each fine grid point carries one scaling function φ^0 and seven "augmented" wavelets $\psi^1 \dots \psi^7$ from the tensor products of φ and ψ along the three axes. (Reproduced from the deck; the curves are the data the demo's `gen_wavelet.py` generates.)

3.4 Recovering φ and ψ: the cascade algorithm

The refinement equation (6) defines φ only implicitly. To plot it (Figure 2) one iterates the two-scale relation starting from a coarse guess, doubling the resolution each pass — the *cascade algorithm*. It converges to φ sampled on a dyadic grid, from which ψ follows by (7). A useful sanity check on the coefficients: $\sum_k h_k = \sqrt{2}$, $\sum_k h_k^2 = 1$, and the m vanishing moments of g_k should be zero to machine precision.

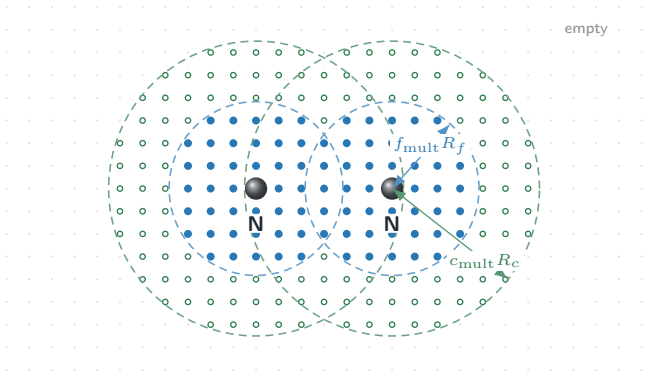
Key idea

Compact support ⇒ locality. A Daubechies orbital is literally zero a few grid spacings away from its centre, so distant orbital pairs do not overlap. That makes the overlap matrix, the Hamiltonian, and the two-electron integral tensor *sparse* — the mathematical reason a wavelet code can scale linearly and the reason ∼94% of the two-electron integrals for a localised system never need to be evaluated (Part C).

4 Part C — Wavelets in BigDFT and computing the integrals

4.1 The adaptive grid

BIGDFT places a Cartesian grid over the molecule and sorts its points into three regions [10]: **fine** points near the atoms carry the full set of eight functions (φ^0 and $\psi^1 \dots \psi^7$); **coarse** points carry only φ^0, i.e. half the resolution; and **empty** points, far from all atoms, carry nothing. The regions are defined by spheres of radius $c_{\text{mult}} R_c$ (coarse) and $f_{\text{mult}} R_f$ (fine) around each atom, with per-species radii from the pseudopotential.



One grid, three regions

- **fine** — ϕ^0 *and* $\psi^{1..7}$
  8 DOF per point ⇒ half the effective spacing
- **coarse** — ϕ^0 only
  1 DOF per point
- **empty** — no basis functions
  vacuum costs nothing

The basis follows the molecule, not a box.
N_2 run: `hgrid 0.25`, `rmult [5.0, 8.0]`
⇒ 130,072 coefficients per orbital

R_c, R_f are per-species radii from the pseudopotential, with $R_f \ll R_c$ — so the fine sphere is the smaller one even though $f_{\text{mult}} = 8.0 > c_{\text{mult}} = 5.0$.

Figure 3: The two-level adaptive grid on a diatomic. The fine region is not a denser mesh — it is the same lattice where each point carries eight basis functions instead of one, i.e. a factor-two resolution increase per dimension. Vacuum costs nothing. (Reproduced from the deck.)

4.2 From grid to orbitals: Kohn–Sham DFT with pseudopotentials

BIGDFT runs Kohn–Sham DFT in this wavelet basis to obtain the occupied and a few virtual MOs. Core electrons are removed by *HGH pseudopotentials* [11] (the relativistic extension of the GTH form [12]), whose external potential splits into a local and a nonlocal (projector) part,

$$V_{\text{ext}} \;\Rightarrow\; V_{\text{psp}} = V_{\text{local}} + \sum_{\ell} \sum_{i,j} |p_i^{(\ell)}\rangle \, h_{ij}^{(\ell)} \, \langle p_j^{(\ell)}|. \tag{8}$$

(The projector coefficients $h_{ij}^{(\ell)}$ here are pseudopotential parameters, unrelated to the integrals h_{pq}/h_{pqrs}.) This split is why the one-electron integral, computed by the project's `toy_model.f90`, comes out in three pieces:

$$h_{pq} = \underbrace{E_{\text{kin}}}_{-\frac{1}{2}\nabla^2} + \underbrace{E_{\text{local}} + E_{\text{nonlocal}}}_{V_{\text{ext}} = V_{\text{psp}}}. \tag{9}$$

Note that h_{pq} is the *wavefunction-theory core* Hamiltonian ($T + V_{\text{ext}}$) — it deliberately excludes the Hartree and exchange–correlation potentials that a KS Hamiltonian would contain, because those belong to the two-electron term.

4.3 The two-electron integral: the Poisson-solver trick

BIGDFT is a DFT code; it has no routine for h_{pqrs}, because Kohn–Sham never needs one. But it *does* solve, every self-consistent iteration, the Hartree problem

$$V_H(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}', \qquad \nabla^2 V_H = -4\pi\rho, \tag{10}$$

with a fast Poisson solver. The kernel is exactly the one h_{pqrs} in (4) requires. The trick is to feed the solver a *transition* density $\rho_{rs} = \psi_r \psi_s$ in place of the physical density, read back the corresponding potential V_{rs}, and contract:

$$h_{pqrs} = \int \rho_{pq}(\mathbf{r})\, V_{rs}(\mathbf{r})\, d\mathbf{r}, \qquad \rho_{pq} = \psi_p^* \psi_q, \quad V_{rs}(\mathbf{r}) = \int \frac{\rho_{rs}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}'. \tag{11}$$

(The BIGDFT orbitals are real, so $\psi_p^* = \psi_p$ here.) A six-dimensional integral collapses to one Poisson solve plus a three-dimensional dot product, reusing a routine BIGDFT already has, tuned and parallel.

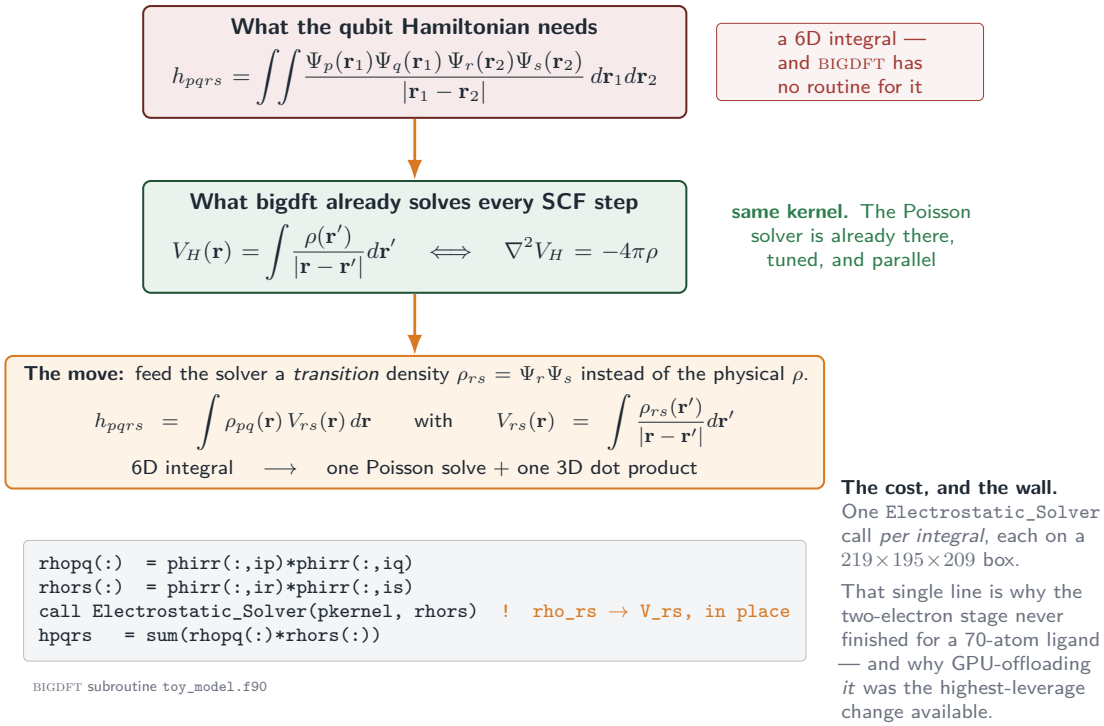


Figure 4: The methodological nugget: obtaining the two-electron integral from BIGDFT's Hartree/Poisson solver by feeding it a transition density. Each integral is one Poisson solve on the real-space grid — which is also why the two-electron stage is the pipeline's bottleneck. (Reproduced from the deck.)

4.4 Symmetries and locality

Two facts cut the work dramatically. First, for real orbitals the chemist integral has eightfold permutation symmetry, $[pq|rs] = [qp|rs] = [pq|sr] = [rs|pq] = \ldots$, and it vanishes unless the spins

match; only the symmetry-unique, spin-allowed entries are computed and the rest reconstructed. Second — and this is where wavelets pay off — if the orbitals are *localised* (Part B), then $\rho_{pq} = \psi_p \psi_q$ is negligible whenever ψ_p and ψ_q sit far apart, so most h_{pqrs} are provably tiny and can be screened out. In the project this removed ∼94% of the two-electron integrals.

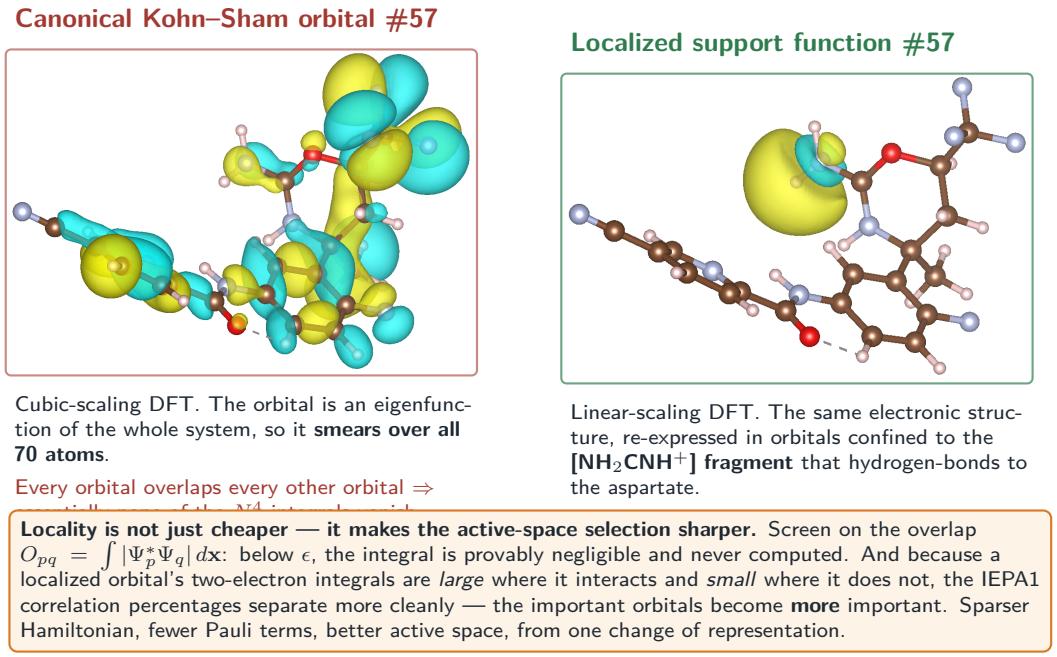


Figure 5: Why locality is the whole game. Left: a canonical Kohn–Sham orbital is delocalised over the entire 70-atom ligand, so every orbital overlaps every other and almost no integral vanishes. Right: the same electronic structure re-expressed in localised support functions collapses onto one fragment; distant pairs stop overlapping and ∼94% of the two-electron integrals are never evaluated. Both are the project's own BIGDFT renders. (Reproduced from the deck.)

The locality that makes support functions cheap is achieved by the linear-scaling formulation of BIGDFT [13, 14]: instead of delocalised canonical orbitals it optimises a minimal set of *localised support functions*, expressed in the same underlying wavelet basis. This rests on *nearsightedness* — the one-particle density matrix $K(\mathbf{r}, \mathbf{r}')$ decays exponentially in $|\mathbf{r} - \mathbf{r}'|$ — so the cost scales linearly rather than cubically with system size.

Tool in our repo

The load + symmetry restoration of h_{pq} and h_{pqrs} is `wavelet-qc-demo/src/wavelet_qc/integrals.py`; rebuilding the Hartree–Fock energy from those integrals to validate them against BIGDFT is `reference.py`. Running `scripts/01_validate_n2.py` rebuilds $E_{\mathrm{HF}} = -19.47$ Ha — the HF energy expression evaluated on the Kohn–Sham orbitals, not a self-consistent HF energy — from the N_2 files and shows it sits the expected +0.44 Ha above BIGDFT's DFT energy −19.91 Ha. On those same orbitals the gap is $E_{xc}^{\mathrm{PBE}}[\rho] - E_x^{\mathrm{exact}}$, i.e. the difference between PBE exchange–correlation and exact HF exchange.

5 Part D — From a Hamiltonian to qubits

5.1 Second quantization

With the integrals in hand, the Hamiltonian is written in *second quantization* on the Fock space of occupation-number states $|n_0 n_1 \dots\rangle$, $n_p \in \{0,1\}$:

$$\hat{H} = \sum_{pq} h_{pq}\, a_p^\dagger a_q + \tfrac{1}{2} \sum_{pqrs} h_{pqrs}\, a_p^\dagger a_r^\dagger a_s a_q, \tag{12}$$

where $a_p^\dagger, a_p$ create and annihilate an electron in spin-orbital p and obey the fermionic anticommutation relations $\{a_p, a_q^\dagger\} = \delta_{pq}$, $\{a_p, a_q\} = 0$. The anticommutators encode the Pauli principle and the antisymmetry of the wavefunction — and they are exactly what makes mapping fermions to qubits nontrivial.

5.2 The mapping problem

A qubit is a distinguishable two-level system; its natural operators (Pauli X, Y, Z) commute on different qubits. Fermionic operators on different modes *anticommute*. A fermion-to-qubit map must reproduce those signs. There are several; all encode the occupation n_p in a qubit, but they differ in how they track the antisymmetry sign.

5.3 Jordan–Wigner

The Jordan–Wigner (JW) transform [15] stores occupation directly — qubit p is $|1\rangle$ iff spin-orbital p is occupied — and attaches a string of Z operators on all lower-indexed qubits to supply the sign:

$$a_p = \Big(\bigotimes_{k<p} Z_k\Big) \frac{X_p + iY_p}{2}, \qquad a_p^\dagger = \Big(\bigotimes_{k<p} Z_k\Big) \frac{X_p - iY_p}{2}. \tag{13}$$

The $\frac{1}{2}(X \pm iY)$ factor flips the target qubit between $|0\rangle$ and $|1\rangle$; the Z-string counts the parity of the electrons below p and produces the ± 1 a fermionic reordering would. Substituting (13) into (12) turns each term into a sum of Pauli strings, giving the *qubit Hamiltonian* $\hat{H}_q = \sum_j c_j P_j$ with $P_j \in \{I, X, Y, Z\}^{\otimes n}$.

Key idea

A worked fragment: the number operator $a_p^\dagger a_p$ maps to $\frac{1}{2}(I - Z_p)$ — eigenvalue 1 on $|1\rangle_p$, 0 on $|0\rangle_p$, exactly the occupation. A hopping term $a_p^\dagger a_q + a_q^\dagger a_p$ (with $p < q$) maps to $\frac{1}{2}(X_p Z_{p+1} \cdots Z_{q-1} X_q + Y_p Z_{p+1} \cdots Z_{q-1} Y_q)$: the Z-string between p and q is the fermionic sign made explicit.

The demo's `mapping.py` builds exactly this operator as a qiskit `SparsePauliOp` and checks it by diagonalising: the ground energy matches an independent PySCF [16] full-CI to $\sim 10^{-14}$ Ha, so the JW signs are provably right.

5.4 Parity, tapering, and Bravyi–Kitaev

JW's Z-strings can be long (up to n qubits), which is awkward on hardware. Two standard refinements:

- **Parity mapping + two-qubit reduction.** Store cumulative parity instead of occupation; the parities of the total electron number and of one spin sector are then carried by two qubits whose values are fixed by those conserved quantities, so they can be removed — a molecule needing n qubits under JW needs $n-2$ under parity [17]. This is the reduction the deck's active-space funnel quotes.
- **Bravyi–Kitaev.** A middle ground between occupation (JW) and parity that stores both in a binary-tree structure, cutting the operator weight to $\mathcal{O}(\log n)$ [18, 19].

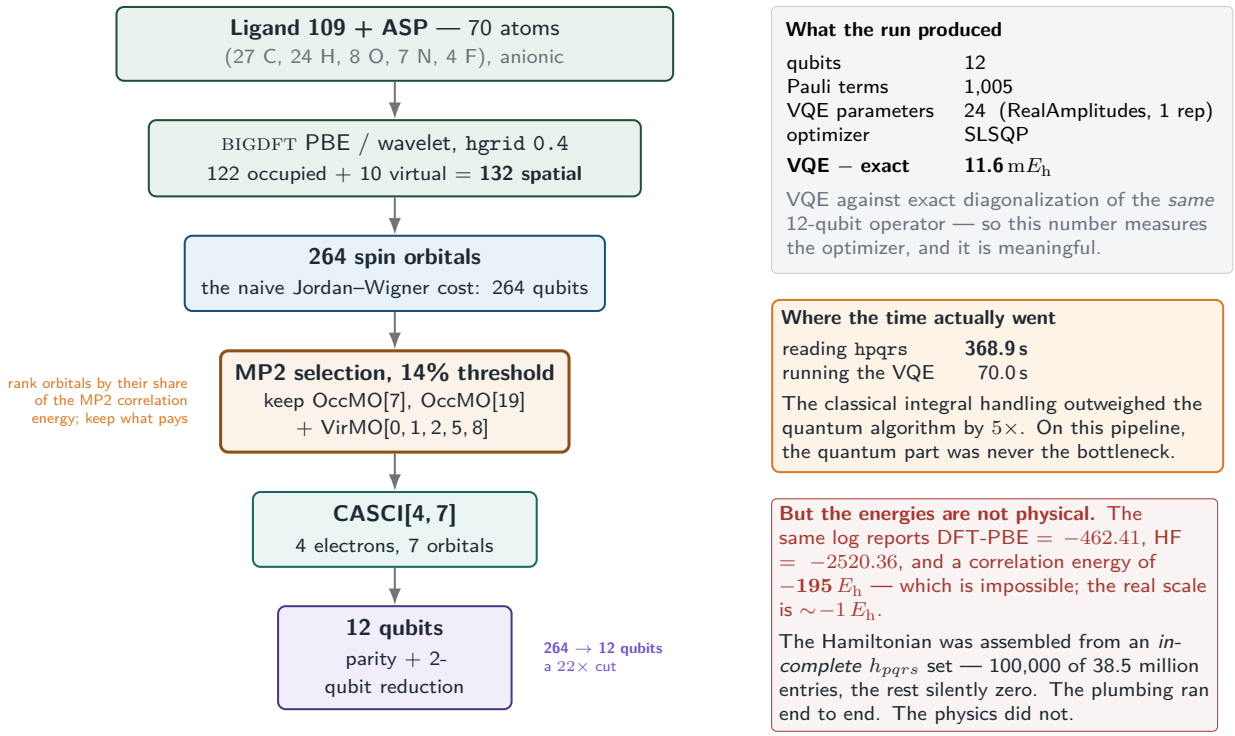


Figure 6: The compression the project actually ran, for the 70-atom ligand: 264 spin orbitals → an MP2-selected CAS[4,7] → parity mapping with a two-qubit reduction → 12 qubits, 1005 Pauli terms. (Reproduced from the deck.)

5.5 Qubit-efficient encodings

JW/parity/BK all use one qubit per spin-orbital. If only a fixed number of electrons m occupy N spin-orbitals, the physical states number only $\binom{N}{m}$, so in principle $Q = \lceil \log_2 \binom{N}{m} \rceil$ qubits suffice — provably fewer than N. The qubit-efficient encoding (QEE) of Shee et al. [20], developed in the same NTU group, realises this: it trades qubits for a larger number of Pauli terms, an explicit resource trade-off rather than a free lunch.

Tool in our repo

The de-interleaving of spin-orbital integrals to spatial blocks and the Jordan–Wigner construction of the `SparsePauliOp` are `wavelet-qc-demo/src/wavelet_qc/mapping.py`. `scripts/03_map_and_diag.py` builds the 8-qubit N_2 Hamiltonian (185 Pauli terms) and confirms its exact diagonalization equals PySCF full-CI.

6 Part E — Solving on a quantum computer

6.1 Active-space selection

Even a minimal wavelet basis gives more orbitals than a near-term device can handle, so one keeps only the orbitals that carry the electron correlation. The criterion used is each orbital's share of the second-order (MP2/IEPA1) correlation energy: occupied and virtual orbitals below a threshold are frozen or removed, leaving a compact active space of n electrons in m orbitals. For N_2 the triple bond dominates — the two degenerate π orbitals and their π^* partners — so a CAS[4,4] or CAS[6,6] captures the essential physics from 15 orbitals.

6.2 The variational quantum eigensolver

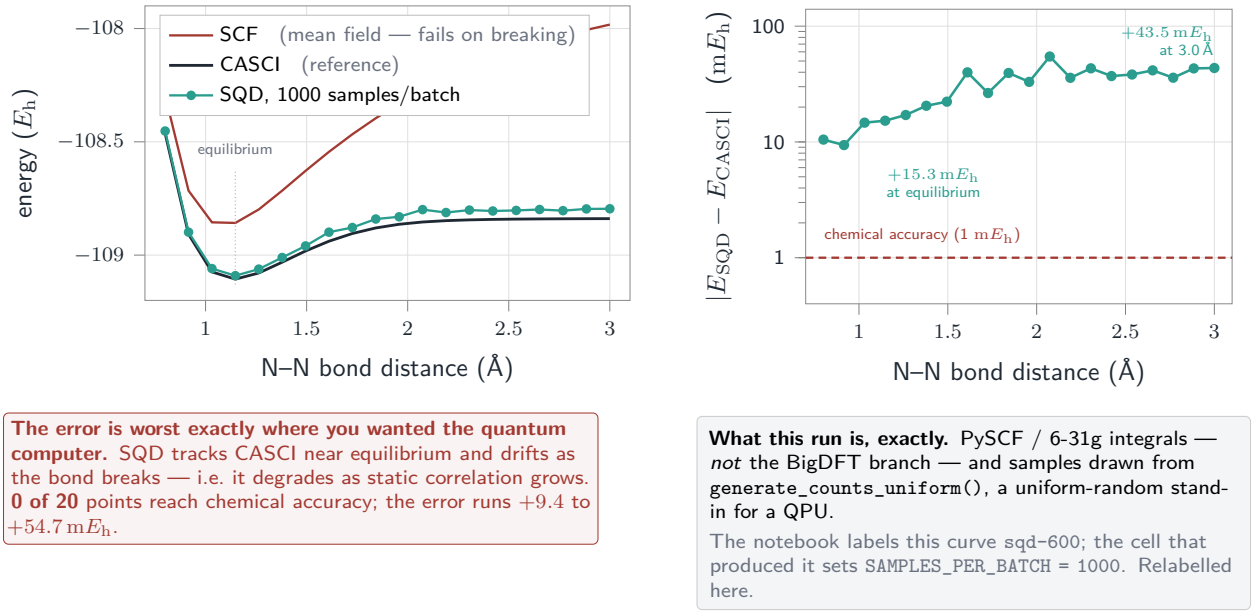
VQE [21, 22] exploits the variational principle $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{H}_q|\psi(\boldsymbol{\theta})\rangle \geq E_0$: prepare a parameterised trial state $|\psi(\boldsymbol{\theta})\rangle$ on the quantum computer, measure its energy, and let a classical optimiser lower $\boldsymbol{\theta}$. Two ansatz families recur:

- **UCCSD** [23], chemistry-inspired — exponentiated single and double excitations of the HF reference; accurate but deep.
- **Hardware-efficient** (e.g. RealAmplitudes) — shallow layers of rotations and entanglers matched to the device; cheap but expressibility-limited, and it plateaus a few mHa above the exact answer.

The demo's `vqe.py` reproduces the second behaviour: on the 8-qubit N_2 active space a hardware-efficient ansatz reaches $\sim$4 mHa of the exact ground state — NISQ-realistic, and consistent with the deck's 11.6 mHa on the larger ligand.

6.3 QSCI / SQD

Sample-based quantum diagonalization (SQD), also called quantum-selected configuration interaction (QSCI) [24, 25], takes a different route: *sample* electron configurations by measuring a state prepared on the quantum computer, then classically *diagonalise* $\hat{H}$ in the subspace those configurations span, refining the subspace over a few "configuration recovery" iterations. The quantum computer proposes a compact set of important determinants; the classical computer does the linear algebra. SQD is systematically improvable (grow the subspace) but has a real coverage limit — amplitude sampling captures the dominant configurations and misses the small-weight tail, so it can plateau above chemical accuracy for strongly correlated systems [26].



The error is worst exactly where you wanted the quantum computer. SQD tracks CASCI near equilibrium and drifts as the bond breaks — i.e. it degrades as static correlation grows. **0 of 20** points reach chemical accuracy; the error runs +9.4 to +54.7 mE_h.

What this run is, exactly. PySCF / 6-31g integrals — *not* the BigDFT branch — and samples drawn from `generate_counts_uniform()`, a uniform-random stand-in for a QPU.

The notebook labels this curve sqd-600; the cell that produced it sets `SAMPLES_PER_BATCH = 1000`. Relabelled here.

Figure 7: SQD on N_2 dissociation: it tracks the reference near equilibrium and degrades as the bond breaks (static correlation grows). The error plateaus above chemical accuracy — the coverage limit above. (Reproduced from the deck; the curve is real stored data, not a sketch.)

The demo reproduces both regimes on the wavelet integrals: on the small CAS[4,4] SQD reaches full-CI exactly (the full determinant space is sampled), while on CAS[6,6] it converges then plateaus a few mHa short — the same trade-off, made runnable in `scripts/05_sqd.py`.

What to read next

For the measurement and low-rank machinery that makes these algorithms practical — factorising the two-electron tensor to cut the number of measurements — see Huggins et al. [27] and Motta et al. [28], both of which the NTU project kept on hand.

7 Making the wavelet method comparable

This is the crucial step, and the one most easily got wrong. The wavelet calculation is *valence-only*: the HGH pseudopotential removes the four 1*s* core electrons and treats only the ten valence electrons, referencing the energy to the pseudopotential. So its total energy ($\sim -20\,E_h$) is not the all-electron total ($\sim -109\,E_h$) with the core missing — the two sit on *different energy zeros*, and the $\sim 89\,E_h$ gap is the frozen core plus an arbitrary pseudopotential shift with no physical meaning as a number. **You must never compare the absolute totals.**

Common confusion

$-19.9\,E_h$ (wavelet, valence) and $-109\,E_h$ (all-electron) differ by the core and a gauge shift, not by a physical error. Subtracting one from the other — or quoting either as "the" energy of N_2 — is meaningless. This is the same reason a plane-wave code never quotes a total energy you would compare to a Gaussian code: you compare cohesive energies, geometries, and vibrations.

7.1 What is comparable

Three classes of quantity survive the change of energy zero:

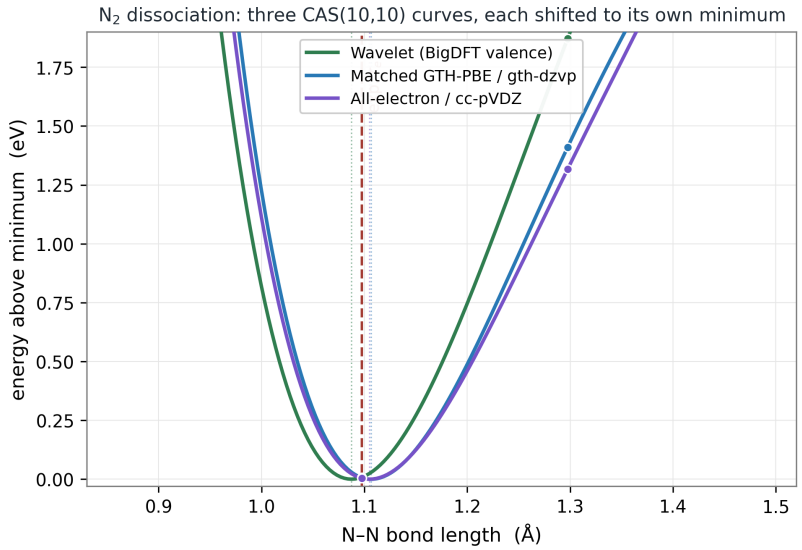
1. **Zero-independent spectroscopic constants.** The equilibrium bond length r_e and the harmonic frequency ω_e come from the *shape* of the potential-energy curve, not its absolute height, so they are directly comparable to any method and to experiment. (This is exactly why Chou et al. [2] benchmarked vibrational *frequencies*: it let a valence pseudopotential wavelet basis be judged against all-electron experiment on equal terms. The deck's 41 cm^{-1} wavelet-vs-cc-pVDZ RMSD is a frequency comparison; no total energies entered it.)
2. **Energy differences in which the core cancels.** Atomization energies $D_e = 2E(\mathrm{N}) - E(\mathrm{N}_2)$, reaction energies, ionization potentials — computed consistently within one method, the (frozen, transferable) core contribution cancels.
3. **Valence correlation energy**, compared to a *frozen-core* all-electron correlation (not the full all-electron value, which also contains core and core–valence correlation the pseudopotential omits).

7.2 The matched-pseudopotential reference

The cleanest apples-to-apples comparison replaces "all-electron vs. pseudopotential" with "*same* pseudopotential, different basis." Running an all-electron code (PySCF) with the *same* GTH-PBE pseudopotential the wavelet calculation used puts both on the same energy zero and the same ten valence electrons; the residual difference is then *purely the basis set* — Daubechies wavelets vs. Gaussians — which is precisely the quantity the whole project is about.

7.3 The result

Figure 8 applies both ideas to N_2. Three CAS(10,10) dissociation curves — the wavelet curve built from the BigDFT integral files, a matched GTH-PBE Gaussian reference, and an all-electron cc-pVDZ reference — are each shifted to their own minimum so their *shapes* overlay. The extracted constants are compared to experiment in the table.



method	r_e (Å)	ω_e (cm^{-1})
Wavelet (BigDFT)	1.087	2549
Matched GTH-PBE	1.107	2453
All-electron cc-pVDZ	1.105	2358
Experiment	**1.098**	**2359**

Zero-independent constants — comparable across the pseudopotential boundary (see note). r_e agrees to ~1%; ω_e is preliminary (coarse 0.2 Å scan, 5 virtuals).

Wavelet from BigDFT integrals via the demo pipeline; references: PySCF CASCI(10,10).

Figure 8: Headline result. N_2 dissociation curves for the wavelet method (from the BigDFT integrals, via the companion demo pipeline) and two PySCF references, all at the CAS(10,10) level and each shifted to its own minimum. The equilibrium bond length r_e agrees to ~1% across all three methods and with experiment — the zero-independent comparison the valence pseudopotential permits. The harmonic frequency ω_e is *preliminary*: the BigDFT scan has coarse 0.2 Å spacing and only five virtual orbitals, so a curvature is under-resolved (the wavelet well is visibly a little too stiff). The matched and all-electron references, built at the same CAS level, validate the procedure. *These numbers will be updated as finer-grid results come in.*

Key idea

The take-home: to compare a valence pseudopotential wavelet calculation to all-electron chemistry, compare *shapes and differences*, not totals — r_e, ω_e, D_e — and use a matched-pseudopotential reference to isolate the basis-set quality. On N_2, the wavelet method already reproduces the bond length to ~1%; sharpening ω_e needs only a finer geometry scan, not a different method.

8 How this maps to the demo

Every concept above is implemented and validated in the companion `wavelet-qc-demo`, which runs the whole chain from the *existing* N_2 integral files (re-running BIGDFT is infeasible). The table ties the two together.

Primer concept	Demo module / script	What it checks
h_{pq}, h_{pqrs} + symmetry (Part A, C)	`integrals.py`	8-fold + spin restoration; loads 30×30, 30^4
HF from integrals (Part C)	`reference.py`, `01_`	$E_{\mathrm{HF}} = -19.47$ vs $E_{\mathrm{KS}} = -19.91$ Ha
MP2 active space (Part E)	`active_space.py`, `02_`	CAS[4,4] from the π/π^* orbitals
Jordan–Wigner (Part D)	`mapping.py`, `03_`	JW exact-diag = PySCF FCI to 10^{-14}
VQE (Part E)	`vqe.py`, `04_`	hardware-efficient VQE within ~4 mHa
QSCI / SQD (Part E)	`sqd.py`, `05_`	subspace diagonalization → full-CI
Comparability (§7)	`dissociation.py`, `06_`	r_e, ω_e vs matched-PSP + all-electron
The resource wall	`09_109asp_wall.py`	truncated integrals, 38.9 GB tensor

The end-to-end plumbing is shown in Figure 9: two executables plus a Python glue layer, with the electron-integral engine in the middle.

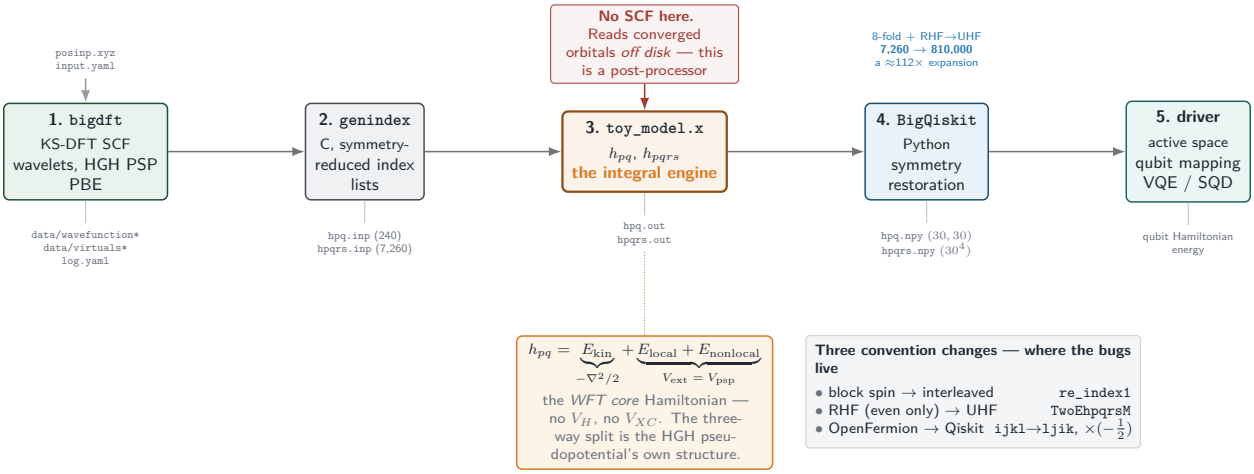


Figure 9: What the code actually does: BIGDFT produces converged wavelet orbitals, a small C tool generates symmetry-reduced index lists, `toy_model.f90` computes the integrals, and a Python layer restores symmetry, selects the active space, maps to qubits, and runs VQE/SQD. (Reproduced from the deck.)

9 Curated reading path

Evening 1 — basis sets and integrals. Szabo & Ostlund [29] chs. 2–3 (Slater determinants, the integrals, HF), with Helgaker et al. [30] as the deeper reference for the second-quantized form; skim the cc-pVXZ idea in Dunning [6].

Evening 2 — wavelets. Mallat [8] for multiresolution analysis; Daubechies [9] for the compactly-supported family; then Genovese et al. [10] for how BIGDFT uses them, and Mohr et al. [13] for the linear-scaling support functions.

Evening 3 — fermions to qubits. Seeley, Richard & Love [19] for JW/BK; Bravyi et al. [17] for tapering; Peruzzo [21] and McClean [22] for VQE; Kanno [24] and Robledo-Moreno [25] for QSCI/SQD.

The NTU line itself. Hong [1] (wavelet MOs), Chou [2] (43 diatomics), Shee [20] (the qubit-efficient encoding) — the three papers this project builds directly on.

10 Glossary

Basis set Finite set of functions used to expand molecular orbitals; its size K sets the (pre-reduction) qubit count.

h_{pq}, h_{pqrs} One- and two-electron integrals; the entire content of the electronic Hamiltonian in a given orbital basis.

Scaling function / wavelet The pair φ, ψ of a multiresolution analysis; coarse averages and fine details, respectively.

Vanishing moments
: Number of polynomial degrees a wavelet integrates to zero against; controls how well the basis represents smooth functions.

Pseudopotential (HGH)
: Effective potential replacing core electrons; its local + nonlocal split gives the $E_{\text{local}} + E_{\text{nonlocal}}$ terms of h_{pq}.

Second quantization
: Operator formalism on Fock space; the Hamiltonian (12) in terms of $a_p^\dagger, a_p$.

Jordan–Wigner
: Fermion-to-qubit map storing occupation with a Z-string for the sign.

Active space
: The subset of orbitals kept for the quantum calculation, chosen here by MP2 correlation share; "CAS[n,m]" = n electrons in m orbitals.

VQE / SQD
: Variational and sample-based quantum eigensolvers; the two routes from a qubit Hamiltonian to an energy.

References

[1] Cheng-Lin Hong et al. "Accurate and Efficient Quantum Computations of Molecular Properties Using Daubechies Wavelet Molecular Orbitals: A Benchmark Study against Experimental Data". In: *PRX Quantum* 3.2 (2022), p. 020360. DOI: 10.1103/PRXQuantum.3.020360.

[2] Shih-Kai Chou, Jyh-Pin Chou, Alice Hu, Yuan-Chung Cheng, and Hsi-Sheng Goan. "Accurate harmonic vibrational frequencies for diatomic molecules via quantum computing". In: *Physical Review Research* 5.4 (2023), p. 043216. DOI: 10.1103/PhysRevResearch.5.043216.

[3] W. J. Hehre, R. F. Stewart, and J. A. Pople. "Self-Consistent Molecular-Orbital Methods. I. Use of Gaussian Expansions of Slater-Type Atomic Orbitals". In: *The Journal of Chemical Physics* 51.6 (1969), pp. 2657–2664. DOI: 10.1063/1.1672392.

[4] R. Ditchfield, W. J. Hehre, and J. A. Pople. "Self-Consistent Molecular-Orbital Methods. IX. An Extended Gaussian-Type Basis for Molecular-Orbital Studies of Organic Molecules". In: *The Journal of Chemical Physics* 54.2 (1971), pp. 724–728. DOI: 10.1063/1.1674902.

[5] W. J. Hehre, R. Ditchfield, and J. A. Pople. "Self-Consistent Molecular Orbital Methods. XII. Further Extensions of Gaussian-Type Basis Sets for Use in Molecular Orbital Studies of Organic Molecules". In: *The Journal of Chemical Physics* 56.5 (1972), pp. 2257–2261. DOI: 10.1063/1.1677527.

[6] Thom H. Dunning. "Gaussian basis sets for use in correlated molecular calculations. I. The atoms boron through neon and hydrogen". In: *The Journal of Chemical Physics* 90.2 (1989), pp. 1007–1023. DOI: 10.1063/1.456153.

[7] Josh J. M. Kirsopp et al. "Quantum computational quantification of protein–ligand interactions". In: *International Journal of Quantum Chemistry* 122.22 (2022), e26975. DOI: 10.1002/qua.26975.

[8] Stéphane G. Mallat. "A Theory for Multiresolution Signal Decomposition: The Wavelet Representation". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11.7 (1989), pp. 674–693. DOI: 10.1109/34.192463.

[9] Ingrid Daubechies. "Orthonormal Bases of Compactly Supported Wavelets". In: *Communications on Pure and Applied Mathematics* 41.7 (1988), pp. 909–996. DOI: 10.1002/cpa.3160410705.

[10] Luigi Genovese et al. "Daubechies wavelets as a basis set for density functional pseudopotential calculations". In: *The Journal of Chemical Physics* 129.1 (2008), p. 014109. DOI: 10.1063/1.2949547.

[11] C. Hartwigsen, S. Goedecker, and J. Hutter. "Relativistic separable dual-space Gaussian pseudopotentials from H to Rn". In: *Physical Review B* 58.7 (1998), pp. 3641–3662. DOI: 10.1103/PhysRevB.58.3641.

[12] S. Goedecker, M. Teter, and J. Hutter. "Separable dual-space Gaussian pseudopotentials". In: *Physical Review B* 54.3 (1996), pp. 1703–1710. DOI: 10.1103/PhysRevB.54.1703.

[13] Stephan Mohr, Laura E. Ratcliff, Paul Boulanger, Luigi Genovese, Damien Caliste, Thierry Deutsch, and Stefan Goedecker. "Daubechies wavelets for linear scaling density functional theory". In: *The Journal of Chemical Physics* 140.20 (2014), p. 204110. DOI: 10.1063/1.4871876.

[14] Stephan Mohr, Laura E. Ratcliff, Luigi Genovese, Damien Caliste, Paul Boulanger, Stefan Goedecker, and Thierry Deutsch. "Accurate and efficient linear scaling DFT calculations with universal applicability". In: *Physical Chemistry Chemical Physics* 17.47 (2015), pp. 31360–31370. DOI: 10.1039/C5CP00437C.

[15] P. Jordan and E. Wigner. "Über das Paulische Äquivalenzverbot". In: *Zeitschrift für Physik* 47.9 (1928), pp. 631–651. DOI: 10.1007/BF01331938.

[16] Qiming Sun et al. "Recent developments in the PySCF program package". In: *The Journal of Chemical Physics* 153.2 (2020), p. 024109. DOI: 10.1063/5.0006074.

[17] Sergey Bravyi, Jay M. Gambetta, Antonio Mezzacapo, and Kristan Temme. *Tapering off qubits to simulate fermionic Hamiltonians*. 2017. arXiv: 1701.08213 [quant-ph].

[18] Sergey B. Bravyi and Alexei Yu. Kitaev. "Fermionic Quantum Computation". In: *Annals of Physics* 298.1 (2002), pp. 210–226. DOI: 10.1006/aphy.2002.6254.

[19] Jacob T. Seeley, Martin J. Richard, and Peter J. Love. "The Bravyi-Kitaev transformation for quantum computation of electronic structure". In: *The Journal of Chemical Physics* 137.22 (2012), p. 224109. DOI: 10.1063/1.4768229.

[20] Yu Shee, Pei-Kai Tsai, Cheng-Lin Hong, Hao-Chung Cheng, and Hsi-Sheng Goan. "Qubit-efficient encoding scheme for quantum simulations of electronic structure". In: *Physical Review Research* 4.2 (2022), p. 023154. DOI: 10.1103/PhysRevResearch.4.023154.

[21] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O'Brien. "A variational eigenvalue solver on a photonic quantum processor". In: *Nature Communications* 5 (2014), p. 4213. DOI: 10.1038/ncomms5213.

[22] Jarrod R. McClean, Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik. "The theory of variational hybrid quantum-classical algorithms". In: *New Journal of Physics* 18.2 (2016), p. 023023. DOI: 10.1088/1367-2630/18/2/023023.

[23] Jonathan Romero, Ryan Babbush, Jarrod R. McClean, Cornelius Hempel, Peter J. Love, and Alán Aspuru-Guzik. "Strategies for quantum computing molecular energies using the unitary coupled cluster ansatz". In: *Quantum Science and Technology* 4.1 (2018), p. 014008. DOI: 10.1088/2058-9565/aad3e4.

[24] Keita Kanno, Masaya Kohda, Ryosuke Imai, Sho Koh, Kosuke Mitarai, Wataru Mizukami, and Yuya O. Nakagawa. *Quantum-Selected Configuration Interaction: classical diagonalization of Hamiltonians in subspaces selected by quantum computers*. 2023. arXiv: 2302.11320 [quant-ph].

[25] Javier Robledo-Moreno, Mario Motta, Holger Haas, Ali Javadi-Abhari, et al. "Chemistry Beyond the Scale of Exact Diagonalization on a Quantum-Centric Supercomputer". In: *Science Advances* 11.25 (2025), eadu9991. DOI: 10.1126/sciadv.adu9991.

[26] Peter Reinholdt, Karl Michael Ziems, Erik Rosendahl Kjellgren, Sonia Coriani, Stephan P. A. Sauer, and Jacob Kongsted. "Critical Limitations in Quantum-Selected Configuration Interaction Methods". In: *Journal of Chemical Theory and Computation* 21.14 (2025), pp. 6811–6822. DOI: 10.1021/acs.jctc.5c00375.

[27] William J. Huggins, Jarrod R. McClean, Nicholas C. Rubin, Zhang Jiang, Nathan Wiebe, K. Birgitta Whaley, and Ryan Babbush. "Efficient and noise resilient measurements for quantum chemistry on near-term quantum computers". In: *npj Quantum Information* 7 (2021), p. 23. DOI: 10.1038/s41534-020-00341-7.

[28] Mario Motta, Erika Ye, Jarrod R. McClean, Zhendong Li, Austin J. Minnich, Ryan Babbush, and Garnet Kin-Lic Chan. "Low-rank representations for quantum simulation of electronic structure". In: *npj Quantum Information* 7 (2021), p. 83. DOI: 10.1038/s41534-021-00416-z.

[29] Attila Szabo and Neil S. Ostlund. *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*. Dover Publications, 1996.

[30] Trygve Helgaker, Poul Jørgensen, and Jeppe Olsen. *Molecular Electronic-Structure Theory*. Wiley, 2000. DOI: 10.1002/9781119019572.